In 2020, how did higher percentages of frequent physical or mental distress moderate the relationship between environmental/ socioeconomic factors and COVID19 cases?

Project One

In [70]:
```
# ! pip install -q pandas
# ! pip install -q pyppeteer
# ! pip install -q qeds
# %pip install numpy
# ! pip install -q pandas
# ! pip install -q pyppeteer
# ! pip install -q qeds
# ! pip install -q matplotlib
# ! pip install -q tabulate
# ! pip install -q statsmodels
# ! pip install -q stargazer
# ! pip install -q qeds fiona geopandas  gensim folium pyLDAvis descartes
#! pip install -q great_tables
import statsmodels.api as sm
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import datetime as dt
from stargazer.stargazer import Stargazer
from IPython.core.display import HTML
from tabulate import tabulate
import geopandas as gpd
from shapely.geometry import Point
import statsmodels.formula.api as smf
from great_tables import GT
from IPython.display import display
import matplotlib.patches as mpatches

%matplotlib inline
# activate plot theme
#import qeds
#qeds.themes.mpl_style();
```

Introduction

- Source of Data

- My dataset comes from a combination of data sources. The socioeconomic factors data comes from the US CDC metrics of "social vulnerability" and natural and man-made disasters. The weather data comes from the NOAA Global Surface Summary of the day, which links each county to a nearby weather station. Also, my dataset uses the County Health Rankings Data a dataset that contains many measures of community health.

- Background
  - Many studies have studied the link between weather and COVID19 cases. The consensus is usually that warmer weather leads to more cases, but the issue is that socioeconomic status could be different in warmer areas and that could be a confounder. For example, Sarkodie et al. (2021) suggests that while air pollution, high temperature, and high humidity can spur COVID-19, underlying health conditions and socioeconomic factors like income can confound this relationship. Additonally, Shen et al. (2020) find a significant effect of limiting socioeconomic activities on COVID-19 cases. Various methods have been used to explore this relationship. For example, Zhang et al. (2022) suggests using Pearson coefficients and Random Forest ML models, and Ahmed et al. (2021) suggests a negative binomial regression model. In my research, I want to affirm that relationship plus explore an additional aspect of consideration known as physical and mental stress. I want to see how this factor interacts with what we already know.
- Research Question
  - In 2020, how did higher percentages of frequent physical or mental distress moderate the relationship between environmental/socioeconomic factors and COVID19 cases across US counties, specifically California?
- Findings
  - Most of the time physical distress has a positive moderating effect on the relationship between socioeconomic/ weather variables and cases. Vice versa for mental distress.
- Citations

  - Benita, F., Rebollar-Ruelas, L., & Gaytán-Alfaro, E. D. (2022). What have we learned about socioeconomic inequalities in the spread of COVID-19? A systematic review. Sustainable cities and society, 86, 104158. https://doi.org/10.1016/j.scs.2022.104158
  - Han, Y., Zhao, W., & Pereira, P. (2022). Global COVID-19 pandemic trends and their relationship with meteorological variables, air pollutants and socioeconomic aspects. Environmental research, 204(Pt C), 112249. https://doi.org/10.1016/j.envres.2021.112249
  - Sarkodie, S. A., & Owusu, P. A. (2021). Global effect of city-to-city air pollution, health conditions, climatic & socio-economic factors on COVID-19 pandemic. The Science of the total environment, 778, 146394. https://doi.org/10.1016/j.scitotenv.2021.146394

  - Ahmed, J., Jaman, M. H., Saha, G., & Ghosh, P. (2021). Effect of environmental and socio-economic factors on the spreading of COVID-19 at 70 cities/provinces. Heliyon, 7(5), e06979. https://doi.org/10.1016/j.heliyon.2021.e06979
  - Zhang, X., Maggioni, V., Houser, P., Xue, Y., & Mei, Y. (2022). The impact of weather condition and social activity on COVID-19 transmission in the United States. Journal of environmental management, 302(Pt B), 114085. https://doi.org/10.1016/j.jenvman.2021.114085
  - Ganslmeier, M., Furceri, D., & Ostry, J. D. (2021). The impact of weather on COVID-19 pandemic. Scientific reports, 11(1), 22027. https://doi.org/10.1038/s41598-021-01189-3
  - Shen, X., Cai, C., & Li, H. (2020). Quantifying socioeconomic activities and weather effects on the global spread of COVID-19 epidemic. https://doi.org/10.48550/arxiv.2005.10223
- Variables
  - X-variables = mean_temp, dewpoint, windspeed, percent_frequent_physical_distress, percent_frequent_mental_distress, percent_no_highschool_diploma, average_daily_pm2_5, unemployment rate, high_school_graduation_rate, percent_limited_access_to_healthy_foods, personal income, real_gdp_2020
  - Y-variable = Cases
- Rationale
  - I chose these variables because of their possible connection to COVID19 cases. Some studies like Ganslmeier et al. (2021) suggest a negative correlation between temperature, humidity, windspeed to COVID19 cases and I wish to explore that correlation with my temperature, dewpoint, and wind speed variables. Also, since Han et al. (2020) mention that the interaction of different types of pollution has a strong impact on cases, I included the PM2.5 variable to explore that effect. The physical and mental distress variables are there to add an additional vector for exploration. I wonder how those variables can interact with the already explored effects of other variables on COVID19 cases. Then I included some socioeconomic variables to cover their potential effect on COVID19 cases. For example, Benita et al. (2022) concludes that income inequality is positively correlated with transmission of the virus. I included education variables, measures of personal income, GDP, unemployment rate as socioeconomic variables. I think it is valuable to explore this since many research have stated there are limitations reducing the accuracy of their research.

Data Cleaning and Loading

In [71]:
```
# Load datasets and do some filtering
```

```python
# Load main data and filter for California only
covid_health_weather = pd.read_csv("C:/Users/Pikachu/OneDrive - University of Toron
covid_health_weather = covid_health_weather[covid_health_weather['state'] == 'Calif

# Get the number of rows and columns in the dataframe
num_rows, num_columns = covid_health_weather.shape
print(f'The dataframe has {num_rows} rows and {num_columns} columns.')


# Load the data for California GDP and Income and filter for 2020 only
# Remove all year columns except for 2020
ca_gdp_2020 = pd.read_csv("c:/Users/Pikachu/OneDrive - University of Toronto/W 2024
ca_gdp_2020 = ca_gdp_2020[['GeoFIPS', 'GeoName', 'Region', 'TableName', 'LineCode',

ca_inc_2020 = pd.read_csv("c:/Users/Pikachu/OneDrive - University of Toronto/W 2024
ca_inc_2020 = ca_inc_2020[['GeoFIPS', 'GeoName', 'Region', 'TableName', 'LineCode',

ca_unemp_2020 = pd.read_csv("c:/Users/Pikachu/OneDrive - University of Toronto/W 20
ca_unemp_2020 = ca_unemp_2020[['FIPS', 'Name', '2020']]

ca_gdp_2020 = ca_gdp_2020[ca_gdp_2020['LineCode'] == 1]
ca_pop_2020 = ca_inc_2020[ca_inc_2020['LineCode'] == 2]
ca_inc_2020 = ca_inc_2020[ca_inc_2020['LineCode'] == 3]
# Rename the 2020 column to real_gdp_2020
ca_gdp_2020.rename(columns = {'2020': 'real_gdp_2020'}, inplace = True)
# Rename to 2020 column to personal_income_capita_2020
ca_inc_2020.rename(columns = {'2020': 'personal_income_capita_2020'}, inplace = Tru
# Rename the 2020 column to unemployment_rate_2020
ca_unemp_2020.rename(columns = {'2020': 'unemployment_rate_2020'}, inplace = True)

ca_pop_2020.rename(columns = {'2020': 'population_2020'}, inplace = True)

# Convert 'FIPS' column to string and remove any extra characters
ca_unemp_2020['FIPS'] = ca_unemp_2020['FIPS'].astype(str).str.replace("'", '').str.
ca_gdp_2020['GeoFIPS'] = ca_gdp_2020['GeoFIPS'].astype(str).str.replace('"', '').st
ca_inc_2020['GeoFIPS'] = ca_inc_2020['GeoFIPS'].astype(str).str.replace('"', '').st
ca_pop_2020['GeoFIPS'] = ca_pop_2020['GeoFIPS'].astype(str).str.replace('"', '').st

# Convert 'fips' column to string to ensure the keys are of the same type
covid_health_weather['fips'] = covid_health_weather['fips'].astype(str).str.replace

# Merge the new datasets into the main dataset using inner join
covid_health_weather = covid_health_weather.merge(ca_gdp_2020, left_on='fips', righ
covid_health_weather = covid_health_weather.merge(ca_pop_2020, left_on='fips', righ
covid_health_weather = covid_health_weather.merge(ca_unemp_2020, left_on='fips', ri
covid_health_weather = covid_health_weather.merge(ca_inc_2020, left_on='fips', righ

# Create a gdp per capita column
covid_health_weather['real_gdp_per_capita_2020'] = covid_health_weather['real_gdp_2

# Change date column to datetime
covid_health_weather['date'] = pd.to_datetime(covid_health_weather['date'])
# Reduce the dataframe to the specified columns
covid_health_weather_reduced = covid_health_weather[['mean_temp', 'dewpoint', 'wind
```

```
The dataframe has 15226 rows and 227 columns.
```

In [72]:

```python
# Creating dataframe with unique values per county


# Create a new dataset that lists the number of cases for each county in 2020
cases_per_county_2020 = covid_health_weather.groupby('county')[['cases']].max().res
cases_per_county_2020.rename(columns={'cases': 'total_cases_2020'}, inplace=True)
#print(cases_per_county_2020)


# Create a new dataset that calculates the average of each variable for each county
avg_per_county = covid_health_weather.groupby('county')[
    [
        'real_gdp_per_capita_2020',
        'unemployment_rate_2020',
        'percent_frequent_physical_distress',
        'percent_frequent_mental_distress',
        'percent_no_highschool_diploma',
        'high_school_graduation_rate',
        'percent_limited_access_to_healthy_foods',
        'personal_income_capita_2020'
    ]
].mean().reset_index()
avg_per_county.columns = ['avg_' + col if col not in ['county'] else col for col in
#print(avg_per_county)

# Merge the two datasets cases_per_county_2020 and avg_per_county
depaneled_data = avg_per_county.merge(cases_per_county_2020, on='county', how='oute
# Add lat and lon to dataset
depaneled_data_geo = depaneled_data.merge(covid_health_weather[['county', 'lat', 'l

# Create total cases per capita column
depaneled_data_geo['total_cases_per_capita'] = depaneled_data_geo['total_cases_2020

# create an index of distress
depaneled_data_geo['distress_index'] = (
    depaneled_data_geo['avg_percent_frequent_physical_distress'] +
    depaneled_data_geo['avg_percent_frequent_mental_distress'] +
    depaneled_data_geo['avg_unemployment_rate_2020']
)
```

Summary Statistics

In [73]:

```python
# Generate summary statistics for the dataset
summary_stats = covid_health_weather[['cases', 'mean_temp', 'real_gdp_2020', 'perso
                                       'percent_frequent_physical_distress', 'perce
                                         'average_daily_pm2_5', 'high_school_gradua
                                           'percent_limited_access_to_healthy_foods
summary_stats = summary_stats.transpose()
summary_stats = summary_stats.round(2)

# Capitalize and remove underscore from index column
summary_stats.index = summary_stats.index.str.replace('_', ' ').str.title()
```

```python
#Remove 25%, 50%, 75%, and max columns from summary_stats
summary_stats_half1 = summary_stats.drop(columns=['25%', '50%', '75%', 'max'])
# Format the new table
display(summary_stats_half1)

# Remove count, mean, std, and min columns from summary_stats
summary_stats_half2 = summary_stats.drop(columns=['count', 'mean', 'std', 'min'])

display(summary_stats_half2)
```

	count	mean	std	min
Cases	15226.0	8284.85	2.903655e+04	1.00
Mean Temp	14250.0	66.03	1.287000e+01	14.50
Real Gdp 2020	15227.0	57187227.54	1.241237e+08	91114.00
Personal Income Capita 2020	15227.0	63159.12	2.457438e+04	33262.00
Dewpoint	13213.0	44.34	1.067000e+01	-9.30
Wind Speed	13740.0	5.27	2.820000e+00	0.00
Percent Frequent Physical Distress	15226.0	11.20	1.540000e+00	8.34
Percent Frequent Mental Distress	15226.0	12.07	1.350000e+00	8.71
Percent No Highschool Diploma	15226.0	16.44	7.200000e+00	5.80
Average Daily Pm2 5	15226.0	10.24	2.850000e+00	5.60
High School Graduation Rate	14976.0	82.48	1.049000e+01	36.30
Unemployment Rate 2020	15227.0	9.90	2.470000e+00	6.80
Percent Limited Access To Healthy Foods	15226.0	5.92	4.380000e+00	0.32

	25%	50%	75%	max
Cases	57.00	621.00	5397.75	4.307130e+05
Mean Temp	57.20	65.85	74.60	1.096000e+02
Real Gdp 2020	2026350.00	9115222.00	44416463.00	2.933320e+09
Personal Income Capita 2020	47615.00	56163.00	68907.00	1.453930e+05
Dewpoint	38.30	46.30	52.40	7.610000e+01
Wind Speed	3.40	4.70	6.50	2.590000e+01
Percent Frequent Physical Distress	10.10	11.06	12.42	1.530000e+01
Percent Frequent Mental Distress	11.27	11.97	13.25	1.504000e+01
Percent No Highschool Diploma	11.00	13.70	21.20	3.300000e+01
Average Daily Pm2 5	8.70	9.60	11.00	1.970000e+01
High School Graduation Rate	81.37	85.24	87.43	1.000000e+02
Unemployment Rate 2020	8.30	9.40	11.10	2.260000e+01
Percent Limited Access To Healthy Foods	3.02	4.77	8.00	2.785000e+01

- Mean_temp is mean temperature. The median appears to be around 18.3 degrees celsius or 65 degrees fahrenheit, but temperature can vary between 14.5 F and 109.6 F. We will connect these varying temps to cases through analysis
- cases counts number of covid19 cases for that city on that day. This will be our main response variable
- real gdp in 2020 is expressed in terms of 2017 dollars. It is one of the key statistics in measuring economic wellbeing of a county and will be one our socioeconomic variables that we are connecting to covid19 cases. GDP could be negatively correlated with cases
- We use personal income to measure income as a socioeconomic variable that can connect to number of cases. Lower income could lead to more cases
- Dewpoint is the temperature the air needs to be cooled to reach a relative humidity of 100%. This is a measure of the moisture in the air. We will use this to predict cases
- Percent frequent and mental distress potentially interacts with cases. We will explore how higher levels of distress impact cases. We will explore how this number interacts with the other factors
- High school graduation rate, umemployment rate, access to healthy goods are some rate based economic variables we will explore on how they are correlated with cases

In [74]:
```
# Define thresholds for high physical and mental distress
high_physical_distress_threshold = covid_health_weather['percent_frequent_physical_
high_mental_distress_threshold = covid_health_weather['percent_frequent_mental_dist
```

```
# Filter the data for high physical and mental distress
high_distress_data = covid_health_weather[
    (covid_health_weather['percent_frequent_physical_distress'] >= high_physical_di
    (covid_health_weather['percent_frequent_mental_distress'] >= high_mental_distre
]

summary_stats_hd = high_distress_data[['cases', 'mean_temp', 'real_gdp_2020', 'pers
                                       'percent_frequent_physical_distress', 'perce
                                         'average_daily_pm2_5', 'high_school_gradua
                                           'percent_limited_access_to_healthy_foods
summary_stats_hd = summary_stats_hd.transpose()
summary_stats_hd = summary_stats_hd.round(2)
# Capitalize and remove underscore from index column
summary_stats_hd.index = summary_stats_hd.index.str.replace('_', ' ').str.title()

#Remove 25%, 50%, 75%, and max columns from summary_stats
summary_stats_half1 = summary_stats_hd.drop(columns=['25%', '50%', '75%', 'max'])
# Format the new table
display(summary_stats_half1)

# Remove count, mean, std, and min columns from summary_stats
summary_stats_half2 = summary_stats_hd.drop(columns=['count', 'mean', 'std', 'min']

display(summary_stats_half2)
```

	count	mean	std	min
Cases	2774.0	4048.87	7687.30	1.00
Mean Temp	2608.0	67.96	13.87	14.50
Real Gdp 2020	2774.0	9628559.45	12763540.96	107899.00
Personal Income Capita 2020	2774.0	46779.39	9428.51	33262.00
Dewpoint	2118.0	44.65	10.10	-3.90
Wind Speed	2357.0	4.98	2.55	0.20
Percent Frequent Physical Distress	2774.0	13.30	0.83	12.54
Percent Frequent Mental Distress	2774.0	13.78	0.47	13.27
Percent No Highschool Diploma	2774.0	21.48	8.85	8.90
Average Daily Pm2 5	2774.0	11.39	4.40	5.60
High School Graduation Rate	2524.0	83.66	4.05	74.36
Unemployment Rate 2020	2774.0	11.95	3.78	8.20
Percent Limited Access To Healthy Foods	2774.0	9.02	2.33	5.38

	25%	50%	75%	max
Cases	22.00	339.00	4296.00	43860.00
Mean Temp	56.80	68.60	78.90	102.30
Real Gdp 2020	783182.00	6201866.00	9115222.00	45610265.00
Personal Income Capita 2020	43949.00	44409.00	45909.00	74115.00
Dewpoint	38.42	45.90	51.78	76.10
Wind Speed	3.30	4.60	6.00	18.10
Percent Frequent Physical Distress	12.66	12.82	13.74	15.30
Percent Frequent Mental Distress	13.48	13.61	13.91	15.04
Percent No Highschool Diploma	11.70	17.80	31.40	33.00
Average Daily Pm2 5	7.80	8.90	14.00	19.70
High School Graduation Rate	80.98	85.25	86.15	88.48
Unemployment Rate 2020	9.40	11.10	12.80	22.60
Percent Limited Access To Healthy Foods	7.41	8.84	9.89	14.00

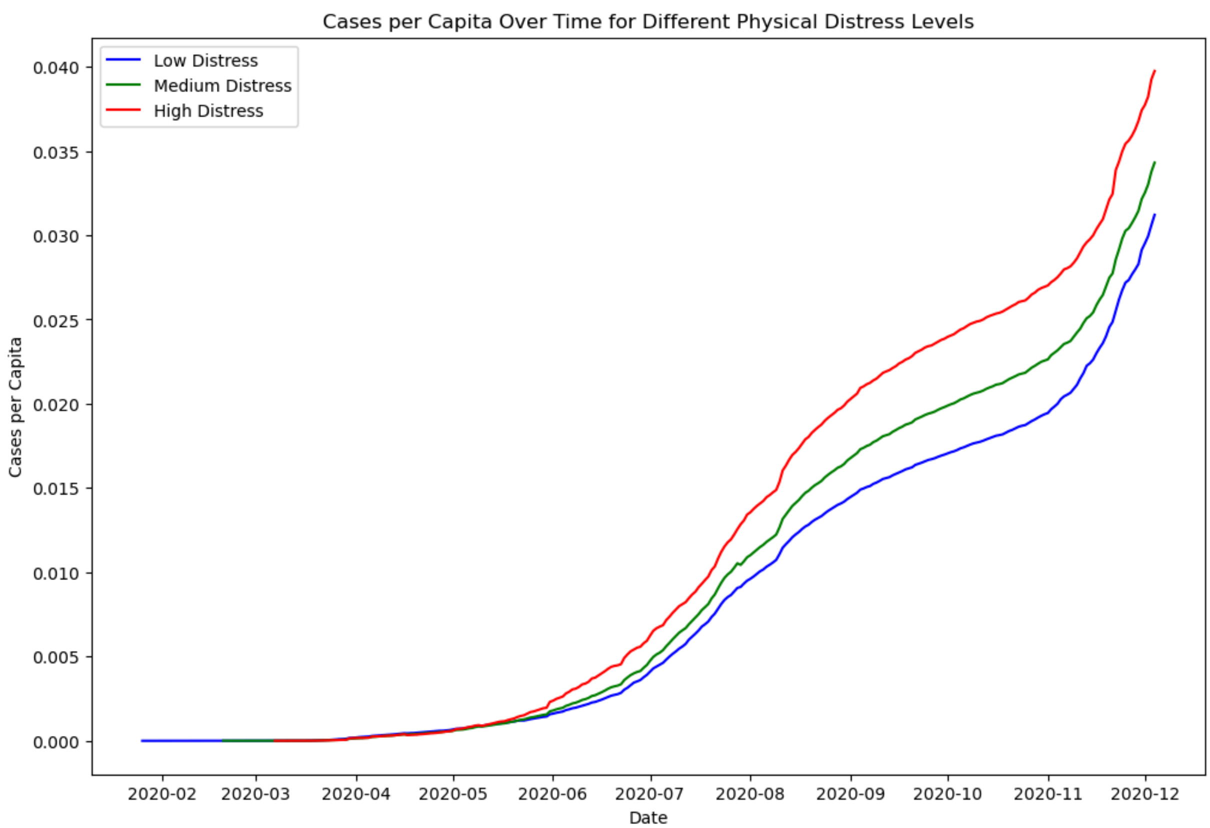
- This table is the summary statistics for the subgroup where distress is higher than the 75th quantile. I will use this subgroup to explore the effect of high distress on cases

Plots

In [75]:

```
# New Plots

# Plot 1 (time series of cases )
# Create a cases by population column in covid_health_weather
covid_health_weather['cases_per_capita'] = covid_health_weather['cases'] / covid_he

# Define distress levels (e.g., using percentiles from your data)
low_physical_distress = covid_health_weather['percent_frequent_physical_distress'].
med_physical_distress = covid_health_weather['percent_frequent_physical_distress'].
high_physical_distress = covid_health_weather['percent_frequent_physical_distress']

# Function to filter data based on physical distress levels
def filter_data(physical_distress_level):
    return covid_health_weather[
        (covid_health_weather['percent_frequent_physical_distress'] >= physical_dis
    ]

# Filter data for different distress levels
low_distress_data = filter_data(low_physical_distress)
med_distress_data = filter_data(med_physical_distress)
high_distress_data = filter_data(high_physical_distress)
```

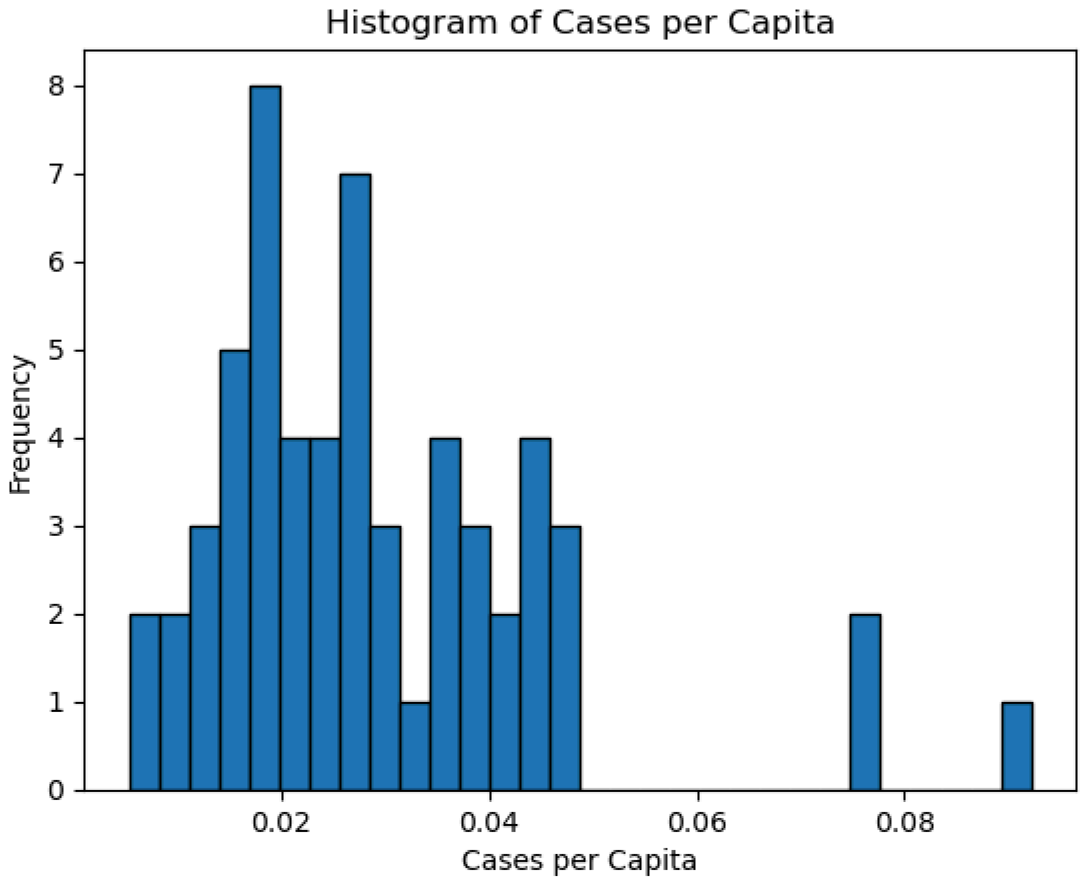
```
# Group by date and calculate the mean for each distress level
low_distress_avg = low_distress_data.groupby('date')['cases_per_capita'].mean()
med_distress_avg = med_distress_data.groupby('date')['cases_per_capita'].mean()
high_distress_avg = high_distress_data.groupby('date')['cases_per_capita'].mean()

# Plot the data
plt.figure(figsize=(12, 8))

plt.plot(low_distress_avg.index, low_distress_avg, label='Low Distress', color='blu
plt.plot(med_distress_avg.index, med_distress_avg, label='Medium Distress', color='
plt.plot(high_distress_avg.index, high_distress_avg, label='High Distress', color='

plt.xlabel('Date')
plt.ylabel('Cases per Capita')
plt.title('Cases per Capita Over Time for Different Physical Distress Levels')
plt.legend()
plt.show()
```



This graph shows how cases evolved over time while grouping by physical distress. It is clear that physical distress has a positive effect on cases

In [76]:
```
# Plot 2 (histrogram of cases per capita)

# Create a histogram of cases per capita
plt.hist(depaneled_data_geo['total_cases_per_capita'], bins=30, edgecolor='black')
plt.title('Histogram of Cases per Capita')
plt.xlabel('Cases per Capita')
```

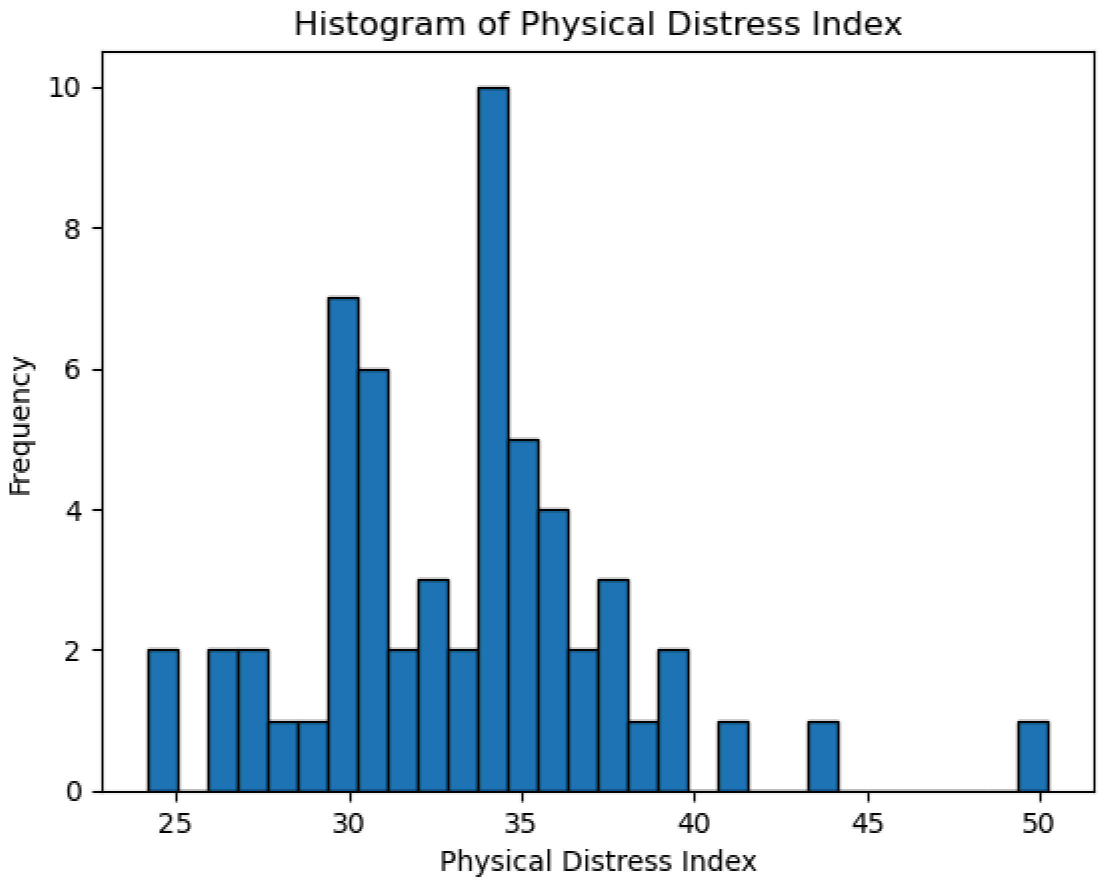
```
plt.ylabel('Frequency')
plt.show()
```



Here we notice a slightly left skewed histogram with some clear high outliers. These counties with high cases per capita are worth additional exploration

In [77]:

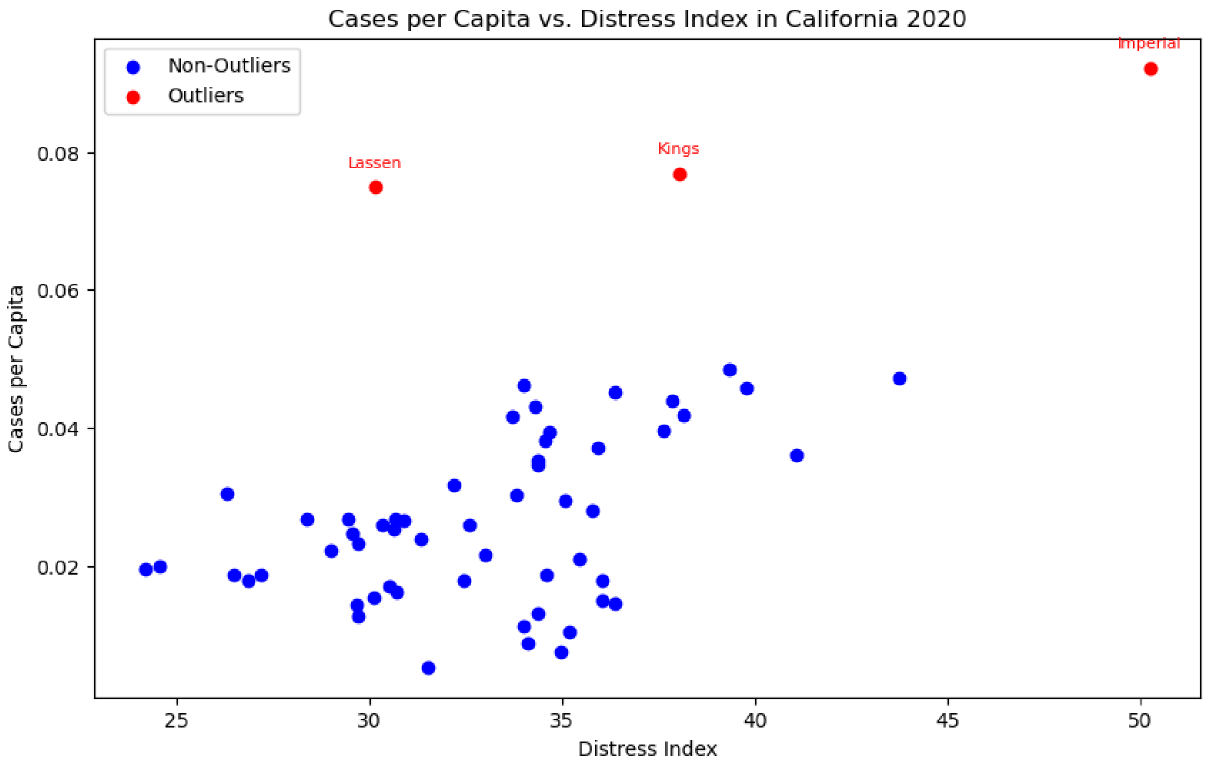
```
# Plot 3 (histogram of distress index)
plt.hist(depaneled_data_geo['distress_index'], bins=30, edgecolor='black')
plt.title('Histogram of Physical Distress Index')
plt.xlabel('Physical Distress Index')
plt.ylabel('Frequency')
plt.show()
```



The distress index of mental, physical distress, and unemployment rate appears to have a fairly non-skewed distribution. This is nice for analysis

In [78]:

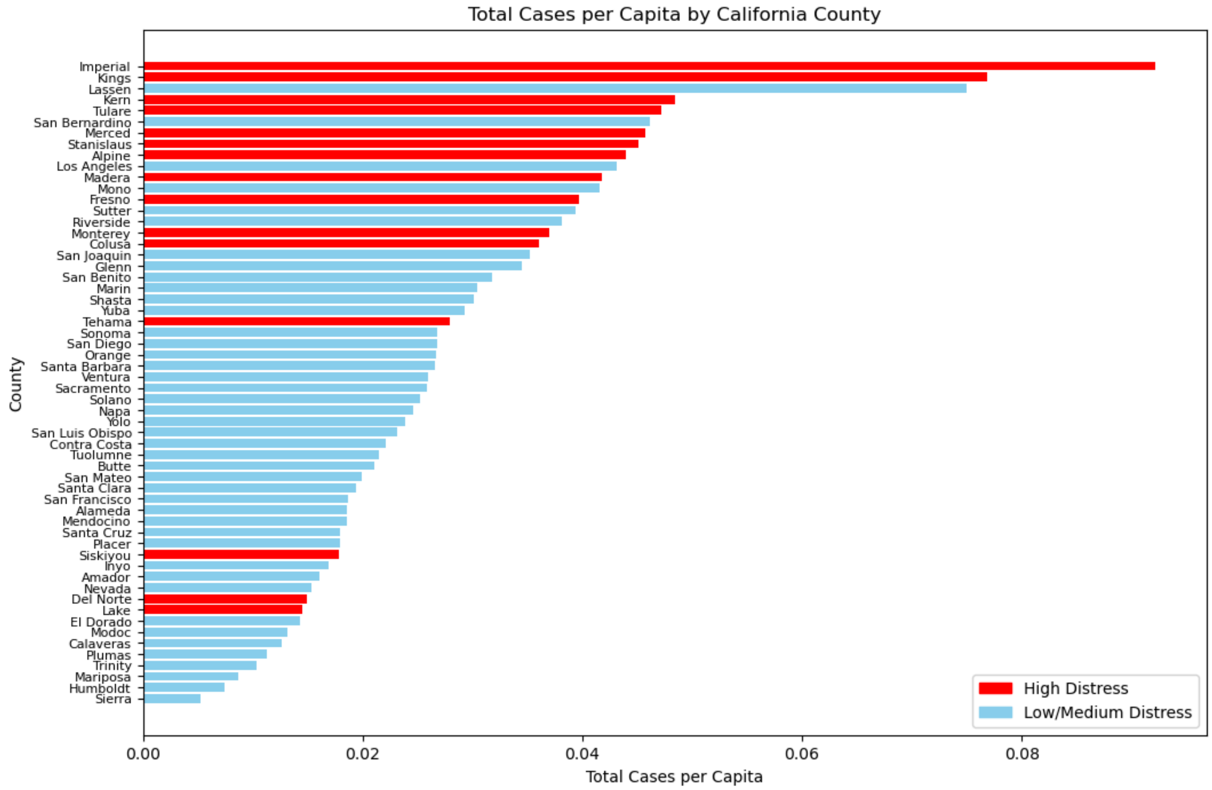
```
# Plot 4 (scatter plot of cases per capita vs. distress index)
plt.figure(figsize=(10, 6))

# Define the threshold for outliers (e.g., 95th percentile)
outlier_threshold = depaneled_data_geo['total_cases_per_capita'].quantile(0.95)

# Plot non-outliers in blue
non_outliers = depaneled_data_geo[depaneled_data_geo['total_cases_per_capita'] <= o
plt.scatter(non_outliers['distress_index'], non_outliers['total_cases_per_capita'],

# Plot outliers in red
outliers = depaneled_data_geo[depaneled_data_geo['total_cases_per_capita'] > outlie
plt.scatter(outliers['distress_index'], outliers['total_cases_per_capita'], color='

# Label the outliers
for i, row in outliers.iterrows():
    plt.annotate(row['county'], (row['distress_index'], row['total_cases_per_capita

plt.title('Cases per Capita vs. Distress Index in California 2020')
plt.xlabel('Distress Index')
plt.ylabel('Cases per Capita')
plt.legend()
plt.show()
```



We see a slight positive correlation between distress and cases.

In [79]:

```
# Plot 5 (bar plot of cases per capita by county)
# Sort the data by total cases per capita
depaneled_data_geo.sort_values('total_cases_per_capita', ascending=False, inplace=T

# Define high distress index threshold (e.g., 75th percentile)
high_distress_threshold = depaneled_data_geo['distress_index'].quantile(0.75)

# Create a color list based on distress index
colors = ['red' if x >= high_distress_threshold else 'skyblue' for x in depaneled_d

# Plot the bar plot
plt.figure(figsize=(12, 8))
plt.barh(depaneled_data_geo['county'], depaneled_data_geo['total_cases_per_capita']
plt.ylabel('County')
plt.xlabel('Total Cases per Capita')
plt.title('Total Cases per Capita by California County')
plt.gca().invert_yaxis()

# Rotate y-axis labels for better readability
plt.yticks(fontsize=8)

# Add legend
red_patch = mpatches.Patch(color='red', label='High Distress')
blue_patch = mpatches.Patch(color='skyblue', label='Low/Medium Distress')
plt.legend(handles=[red_patch, blue_patch])

plt.show()
```



Here I visualize in a bar chart which counties have the greatest number of cases per capita. And highlight whether they have high or low/ medium distress. We see most areas with high distress have higher cases

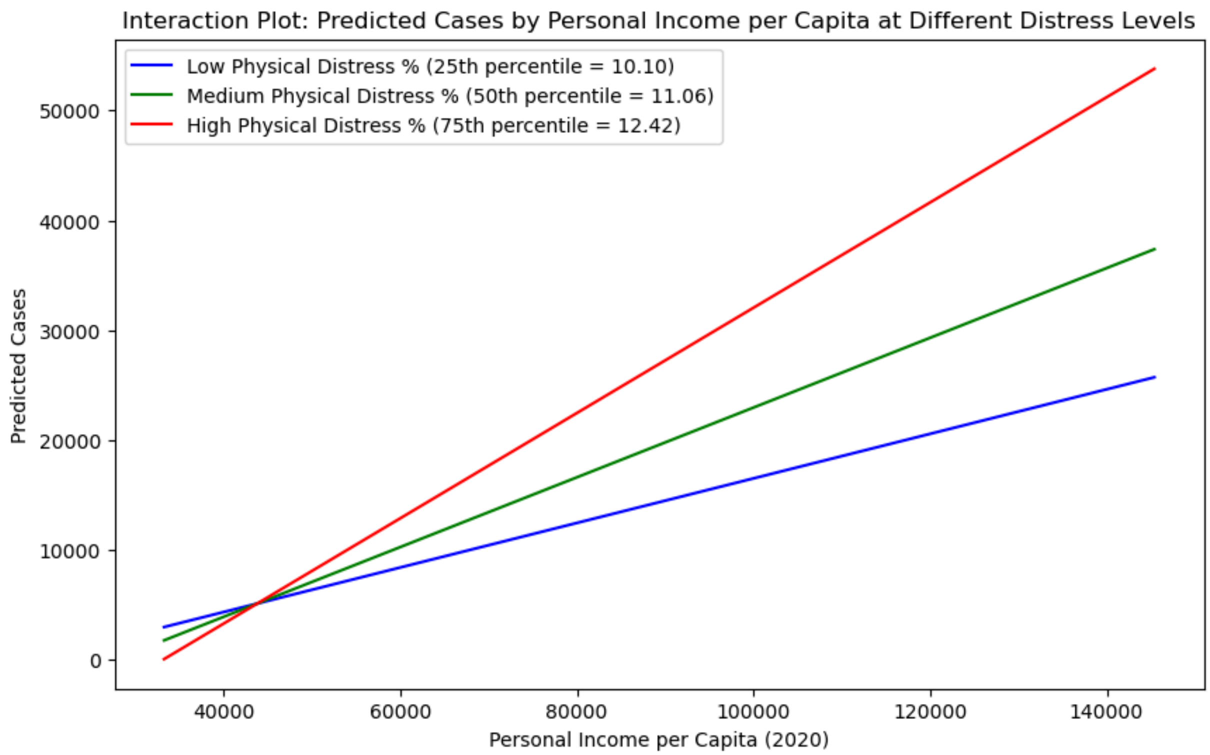
Project 2

Message

The main message is that physical distress has a positive impact on cases and the relationship between cases and other variables.

In [80]:
```
# Fit a regression model including the interaction term
model = smf.ols('cases ~ personal_income_capita_2020 * percent_frequent_physical_di

# Generate a range of values for the environmental/socioeconomic factor
env_range = np.linspace(covid_health_weather['personal_income_capita_2020'].min(),

# Define distress levels (e.g., using percentiles from your data)
low_distress = covid_health_weather['percent_frequent_physical_distress'].quantile(
med_distress = covid_health_weather['percent_frequent_physical_distress'].quantile(
high_distress = covid_health_weather['percent_frequent_physical_distress'].quantile

# Function to predict cases given a distress level
def predict_cases(env_values, distress_level):
    pred_df = pd.DataFrame({
        'personal_income_capita_2020': env_values,
```

```
        'percent_frequent_physical_distress': distress_level
    })
    return model.predict(pred_df)

# Calculate predicted values for each distress level
pred_low = predict_cases(env_range, low_distress)
pred_med = predict_cases(env_range, med_distress)
pred_high = predict_cases(env_range, high_distress)

# Plot the interaction: predicted cases vs. personal income per capita for differen
plt.figure(figsize=(10, 6))
plt.plot(env_range, pred_low, label=f'Low Physical Distress % (25th percentile = {l
plt.plot(env_range, pred_med, label=f'Medium Physical Distress % (50th percentile =
plt.plot(env_range, pred_high, label=f'High Physical Distress % (75th percentile =

plt.xlabel('Personal Income per Capita (2020)')
plt.ylabel('Predicted Cases')
plt.title('Interaction Plot: Predicted Cases by Personal Income per Capita at Diffe
plt.legend()
plt.show()
```



Regression

I think the relationship between my x variables and y variables are linear. First of all, past research regarding topics have used linear type regression. For example, Ahmed et al. (2021) mention that covid-19 case count is skewed so using a log-linear regression model to fix it is appropriate. This is the case for my model since from the summary statistics table it is clear that my response is skewed.

This first regression table attempts to discover the effect of important explanatory variables on my dependent variable cases. The table explores the effect of the main x variables and then adds socioeconomic and weather controls. I'd like to understand the effect of these variables on cases and whether the introduction of one control impacts the effect of another variable. The first regression introduces the effect of the main distress variables. The 2nd regression introduces the important variable of personal income, because past research like Benita. et al (2022) mention this variable has a strong positive association with cases. The 3rd and 4th regression introduce socioeconomic and weather controls respectively.

$$\text{cases} = \beta_0 + \beta_1 \cdot \text{physical distress} + \beta_2 \cdot \text{mental distress} + \beta_3 \cdot \text{socioeconomic and weath}$$

In [81]:

```
# Log transform the dependent variable
y = np.log(covid_health_weather['cases'] + 1)

# Log transform the independent variables
covid_health_weather['log_percent_frequent_physical_distress'] = np.log(covid_healt
covid_health_weather['log_percent_frequent_mental_distress'] = np.log(covid_health_
covid_health_weather['log_personal_income_capita_2020'] = np.log(covid_health_weath
covid_health_weather['log_unemployment_rate_2020'] = np.log(covid_health_weather['u
covid_health_weather['log_real_gdp_per_capita_2020'] = np.log(covid_health_weather[
covid_health_weather['log_mean_temp'] = np.log(covid_health_weather['mean_temp'] +
covid_health_weather['log_average_daily_pm2_5'] = np.log(covid_health_weather['aver

# Regression without controls
X1 = covid_health_weather[['log_percent_frequent_physical_distress', 'log_percent_f
X1 = sm.add_constant(X1)  # Add a constant term to the predictors

model1 = sm.OLS(y, X1, missing='drop').fit()

X2 = covid_health_weather[['log_percent_frequent_physical_distress', 'log_percent_f
X2 = sm.add_constant(X2)  # Add a constant term to the predictors
model2 = sm.OLS(y, X2, missing='drop').fit()

# Regression with some controls
X3 = covid_health_weather[['log_percent_frequent_physical_distress', 'log_percent_f
X3 = sm.add_constant(X3)  # Add a constant term to the predictors
model3 = sm.OLS(y, X3, missing='drop').fit()

# Regression with more controls
X4 = covid_health_weather[['log_percent_frequent_physical_distress', 'log_percent_f
X4 = sm.add_constant(X4)  # Add a constant term to the predictors
model4 = sm.OLS(y, X4, missing='drop').fit()

# Create a Stargazer summary table for all models
stargazer = Stargazer([model1, model2, model3, model4])
stargazer.covariate_order(['const', 'log_percent_frequent_physical_distress', 'log_
HTML(stargazer.render_html())

# # Save the Stargazer summary table to an HTML file
# with open('regression_summary.html', 'w') as f:
#     f.write(stargazer.render_html())
```

Out[81]:

	Dependent variable: cases			
	(1)	(2)	(3)	(4)
const	31.589***	14.421***	5.922***	-19.082***
	(0.522)	(1.993)	(2.266)	(2.186)
log_percent_frequent_physical_distress	23.501***	24.386***	28.853***	16.454***
	(0.433)	(0.443)	(0.604)	(0.599)
log_percent_frequent_mental_distress	-32.722***	-31.243***	-35.517***	-21.932***
	(0.519)	(0.543)	(0.661)	(0.647)
log_personal_income_capita_2020		1.015***	2.346***	2.266***
		(0.114)	(0.187)	(0.173)
log_unemployment_rate_2020			-0.931***	-1.282***
			(0.153)	(0.145)
log_real_gdp_per_capita_2020			-1.028***	-0.739***
			(0.113)	(0.104)
log_mean_temp				2.669***
				(0.100)
log_average_daily_pm2_5				4.371***
				(0.090)
Observations	15226	15226	15226	14250
R^2	0.213	0.217	0.223	0.370
Adjusted R^2	0.212	0.217	0.223	0.370
Residual Std. Error	2.545 (df=15223)	2.538 (df=15222)	2.528 (df=15220)	2.273 (df=14242)
F Statistic	2054.617*** (df=2; 15223)	1403.350*** (df=3; 15222)	874.398*** (df=5; 15220)	1196.569*** (df=7; 14242)
Note:				*p<0.1; **p<0.05; ***p<0.01

Most of my variables have a statistically significant coefficent at the 1% significance level which suggests the model is accurate. Also, the F statistic is high which suggests that the models as a whole are significant. In terms of the regression coefficients, the most important thing to note is that there is a significant positive effect of physical distress on cases and vice versa for mental distress. The positive effect of physical distress on cases follows intuition and the negative effect of mental distress on cases requires more exploration.

2nd regression

This regression adds on to the previous regression and explores the main point of the research question. The moderation effect of distress on the relationship between socioeconomic/ weather variables and cases.

$$\text{cases} = \beta_0 + \beta_1 \cdot \text{previous model} + \beta_2 \cdot \text{distress and socioeconomic/ weather interaction}$$

In [82]:

```
# Regression with more controls (same as model4)
X4 = covid_health_weather[['log_percent_frequent_physical_distress', 'log_percent_f
X4 = sm.add_constant(X4)  # Add a constant term to the predictors
model4 = sm.OLS(y, X4, missing='drop').fit()

# Regression with interaction term: physical distress * personal income
X5 = covid_health_weather[['log_percent_frequent_physical_distress', 'log_percent_f
X5 = X5.assign(physical_distress_income_interaction=covid_health_weather['log_perce
X5 = sm.add_constant(X5)  # Add a constant term to the predictors
model5 = sm.OLS(y, X5, missing='drop').fit()

# Regression with interaction term: mental distress * unemployment rate
X6 = covid_health_weather[['log_percent_frequent_physical_distress', 'log_percent_f
X6 = X6.assign(mental_distress_unemployment_interaction=covid_health_weather['log_p
X6 = sm.add_constant(X6)  # Add a constant term to the predictors
model6 = sm.OLS(y, X6, missing='drop').fit()

# Regression with interaction term: physical distress * mean temperature
X7 = covid_health_weather[['log_percent_frequent_physical_distress', 'log_percent_f
X7 = X7.assign(physical_distress_temp_interaction=covid_health_weather['log_percent
X7 = sm.add_constant(X7)  # Add a constant term to the predictors
model7 = sm.OLS(y, X7, missing='drop').fit()

# Regression with interaction term: mental distress * pm2.5
X8 = covid_health_weather[['log_percent_frequent_physical_distress', 'log_percent_f
X8 = X8.assign(mental_distress_pm25_interaction=covid_health_weather['log_percent_f
X8 = sm.add_constant(X8)  # Add a constant term to the predictors
model8 = sm.OLS(y, X8, missing='drop').fit()

# Create a Stargazer summary table for all models
stargazer = Stargazer([model4, model5, model6, model7])
stargazer.covariate_order(['const', 'log_percent_frequent_physical_distress', 'log_
HTML(stargazer.render_html())
```

Out[82]:

	Dependent variable: cases			
	(1)	(2)	(3)	(4)
const	-19.082***	197.923***	-134.180***	-85.992***
	(2.186)	(14.231)	(8.095)	(8.953)
log_percent_frequent_physical_distress	16.454***	-74.116***	18.590***	42.688***
	(0.599)	(5.900)	(0.612)	(3.456)
log_percent_frequent_mental_distress	-21.932***	-23.370***	16.862***	-21.824***
	(0.647)	(0.648)	(2.706)	(0.645)
log_personal_income_capita_2020	2.266***	-17.730***	3.028***	2.296***
	(0.173)	(1.307)	(0.180)	(0.173)
log_unemployment_rate_2020	-1.282***	-1.886***	42.837***	-1.152***
	(0.145)	(0.149)	(2.993)	(0.146)
log_real_gdp_per_capita_2020	-0.739***	-0.660***	-0.641***	-0.728***
	(0.104)	(0.103)	(0.103)	(0.104)
log_mean_temp	2.669***	2.700***	2.727***	18.457***
	(0.100)	(0.099)	(0.099)	(2.051)
log_average_daily_pm2_5	4.371***	4.829***	4.628***	4.487***
	(0.090)	(0.094)	(0.091)	(0.091)
physical_distress_income_interaction		8.482***		
		(0.550)		
mental_distress_unemployment_interaction			-17.069***	
			(1.157)	
physical_distress_temp_interaction				-6.305***
				(0.818)
Observations	14250	14250	14250	14250
R^2	0.370	0.381	0.380	0.373
Adjusted R^2	0.370	0.380	0.379	0.373
Residual Std. Error	2.273 (df=14242)	2.255 (df=14241)	2.256 (df=14241)	2.269 (df=14241)

F Statistic	1196.569*** (df=7; 14242)	1094.178*** (df=8; 14241)	1090.158*** (df=8; 14241)	1058.712*** (df=8; 14241)
Note:				*p<0.1; **p<0.05; ***p<0.01

In [83]:

```
# Create a Stargazer summary table for model8
stargazer_model8 = Stargazer([model8])
stargazer_model8.covariate_order(['const', 'log_percent_frequent_physical_distress'
HTML(stargazer_model8.render_html())
```

Out[83]:

	Dependent variable: cases
	(1)
const	-62.154***
	(6.735)
log_percent_frequent_physical_distress	17.539***
	(0.619)
log_percent_frequent_mental_distress	-6.788***
	(2.331)
log_personal_income_capita_2020	2.363***
	(0.174)
log_unemployment_rate_2020	-1.308***
	(0.145)
log_real_gdp_per_capita_2020	-0.790***
	(0.104)
log_mean_temp	2.655***
	(0.100)
log_average_daily_pm2_5	22.069***
	(2.619)
mental_distress_pm25_interaction	-6.772***
	(1.002)
Observations	14250
R^2	0.372
Adjusted R^2	0.372
Residual Std. Error	2.270 (df=14241)
F Statistic	1055.996*** (df=8; 14241)
Note:	*p<0.1; **p<0.05; ***p<0.01

Most of my variables have a statistically significant coefficent at the 1% significance level which suggests the model is accurate. Also, the F statistic is high which suggests that the models as a whole are significant. In terms of the regression coefficients, the positive coefficient for the interaction between physical distress and income suggests that as physical distress increases the impact of income on cases increases. However, in terms of the interaction betweeen mental distress and unemployment, the negative coefficient suggests that as mental distress increases the effect of unemployment on cases decreases. The other two interaction terms have a similar negative coefficient leading to a similar interpretation.

For my research question, this suggests that physical distress increasingly affects the relationship between socioeconomic cases plus weather and cases and vice versa for mental distress.

This regression is my preferred specification for its insight into the moderation effect that my research question is asking.

Maps

In [84]:
```
# Prepare the data for mapping

depaneled_data_geo["Coordinates"] = list(zip(depaneled_data_geo.lon, depaneled_data
depaneled_data_geo["Coordinates"] = depaneled_data_geo["Coordinates"].apply(Point)
#covid_health_weather.head()

gdf = gpd.GeoDataFrame(depaneled_data_geo, geometry="Coordinates")
#gdf.head()
```

In [85]:
```
# Query California map borders
state_df = gpd.read_file("http://www2.census.gov/geo/tiger/GENZ2016/shp/cb_2016_us_
state_df.head()

fig, gax = plt.subplots(figsize=(10, 10))
state_df.query("NAME == 'California'").plot(ax=gax, edgecolor="black", color="white
#plt.show()
```

Out[85]: <Axes: >

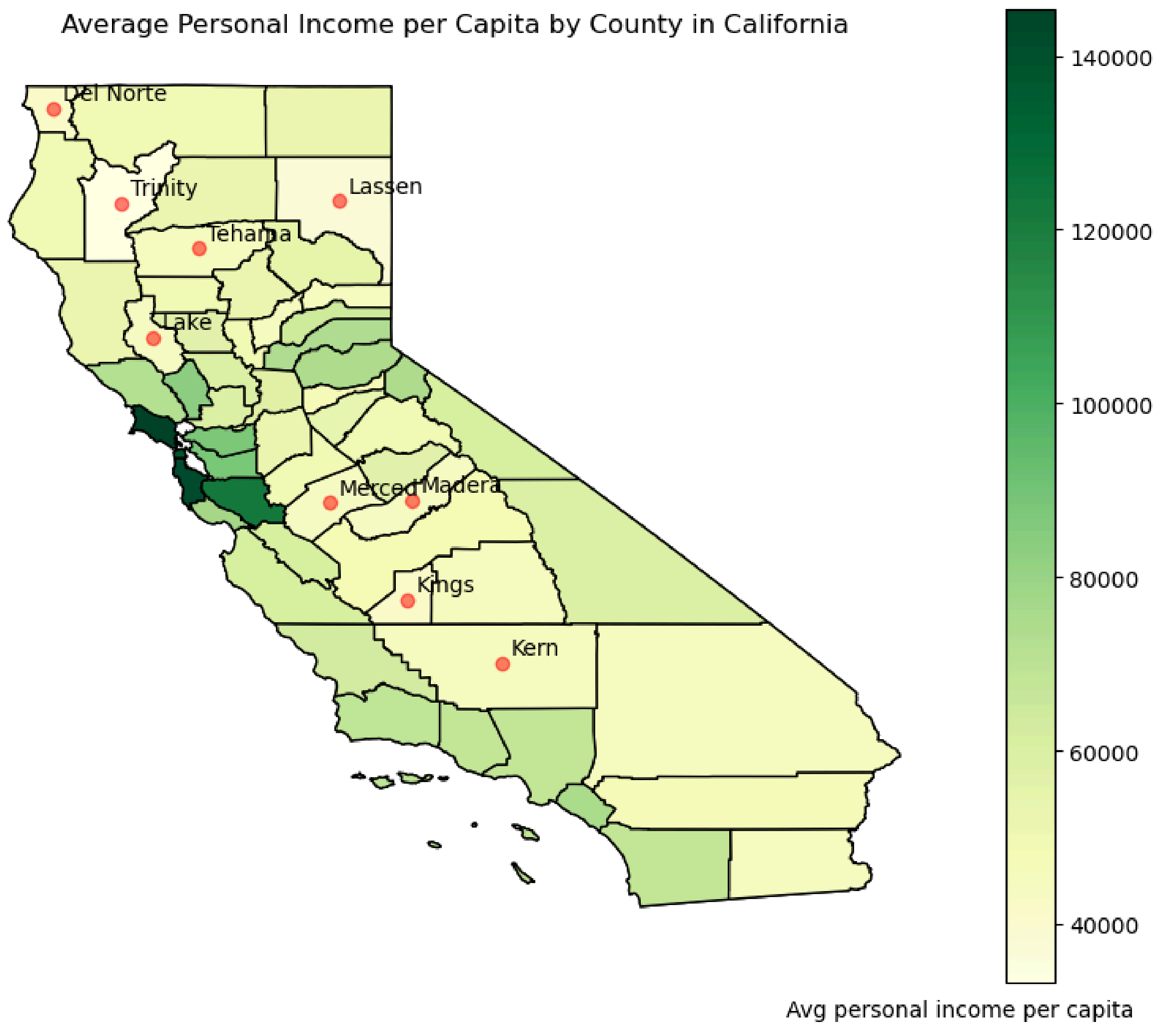
In [86]:
```
# Plot California county borders
county_df = gpd.read_file("http://www2.census.gov/geo/tiger/GENZ2016/shp/cb_2016_us

county_df = county_df.query("STATEFP == '06'")

fig, gax = plt.subplots(figsize=(10, 10))

county_df.plot(ax=gax, edgecolor="black", color="white")

#plt.show()
```

Out[86]: <Axes: >

In [87]:
```
gdf["county"] = gdf["county"].str.title()
gdf["county"] = gdf["county"].str.strip()
```

```
county_df["NAME"] = county_df["NAME"].str.title()
county_df["NAME"] = county_df["NAME"].str.strip()
# merge county data with covid data
county_df_covid = county_df.merge(gdf, left_on="NAME", right_on="county", how="inne
#county_df_covid.head()
```

In [93]:
```
fig, gax = plt.subplots(figsize = (10,8))

# Plot the state level
state_df[state_df['NAME'] == 'California'].plot(ax = gax, edgecolor='black',color='

# Plot the counties and data
county_df_covid.plot(
    ax=gax, edgecolor='black', column='avg_personal_income_capita_2020', legend=Tru
)

# Add text to let people know what we are plotting
gax.annotate('Avg personal income per capita',xy=(0.53, 0.05),  xycoords='figure fr
gdf.head()
# Plot county names of high personal income per capita
gdf_highquantile = gdf[gdf['avg_personal_income_capita_2020'] < gdf['avg_personal_i
gdf_highquantile.plot(ax=gax, color='red', alpha = 0.5)
for x, y, label in zip(gdf_highquantile['Coordinates'].x, gdf_highquantile['Coordin
    gax.annotate(label, xy=(x,y), xytext=(4,4), textcoords='offset points')


# I don't want the axis with long and lat
plt.axis('off')

# Add a title
plt.title('Average Personal Income per Capita by County in California')

plt.show()
```

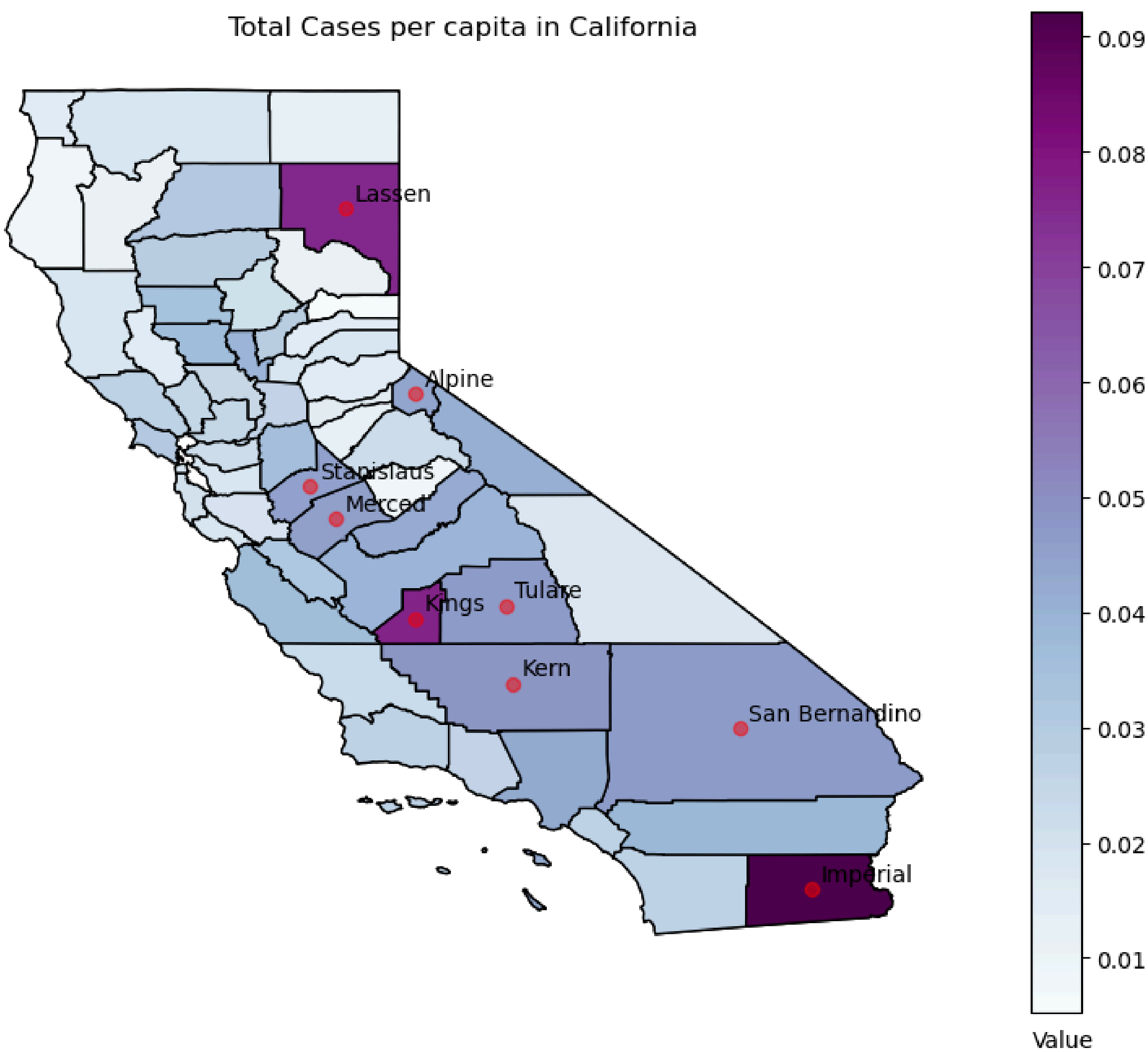


In this map, I have highlighted the average personal income per capita in California. These locations with low average personal income per capita may respond worse to Covid-19

In [89]:

```
fig, gax = plt.subplots(figsize = (10,8))

# Plot the state level
state_df[state_df['NAME'] == 'California'].plot(ax = gax, edgecolor='black',color='

# Plot the counties and data
county_df_covid.plot(
    ax=gax, edgecolor='black', column='total_cases_per_capita', legend=True, cmap='
)

# Add text to let people know what we are plotting
gax.annotate('Value',xy=(0.67, 0.05),  xycoords='figure fraction')
gdf.head()
# Plot county names of high personal income per capita
gdf_highquantile = gdf[gdf['total_cases_per_capita'] > gdf['total_cases_per_capita'
gdf_highquantile.plot(ax=gax, color='red', alpha = 0.5)
for x, y, label in zip(gdf_highquantile['Coordinates'].x, gdf_highquantile['Coordin
    gax.annotate(label, xy=(x,y), xytext=(4,4), textcoords='offset points')
```

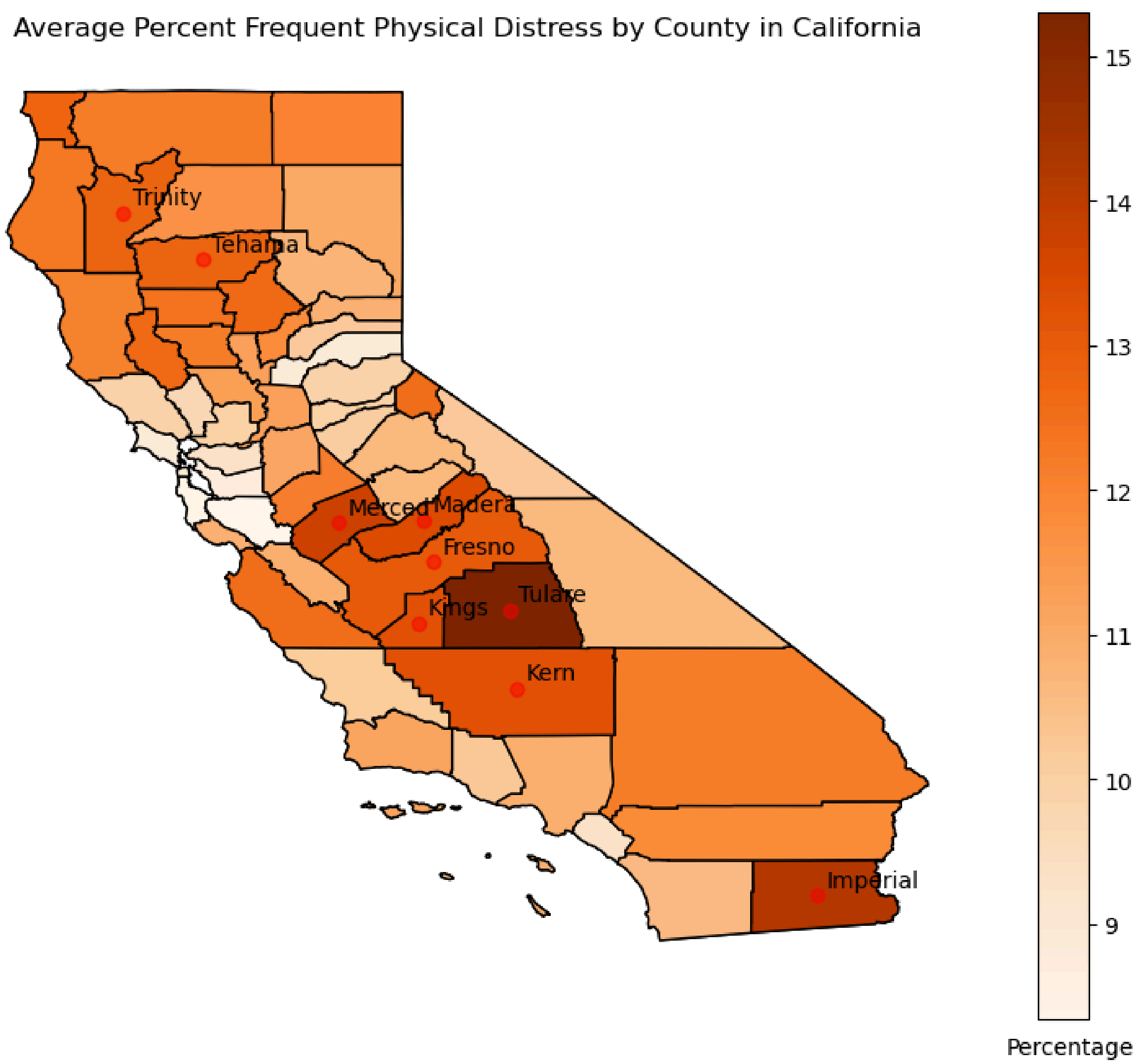
```
# I don't want the axis with long and lat
plt.axis('off')

# Add a title
plt.title('Total Cases per capita in California')

plt.show()
```



This is a map highlighting the main response variable cases per capita. I highlight the areas with the most cases per capita. We will look for correlation between the x variables and cases.

In [90]:
```
fig, gax = plt.subplots(figsize = (10,8))

# Plot the state level
state_df[state_df['NAME'] == 'California'].plot(ax = gax, edgecolor='black',color='

# Plot the counties and data
county_df_covid.plot(
    ax=gax, edgecolor='black', column='avg_percent_frequent_physical_distress', leg
```

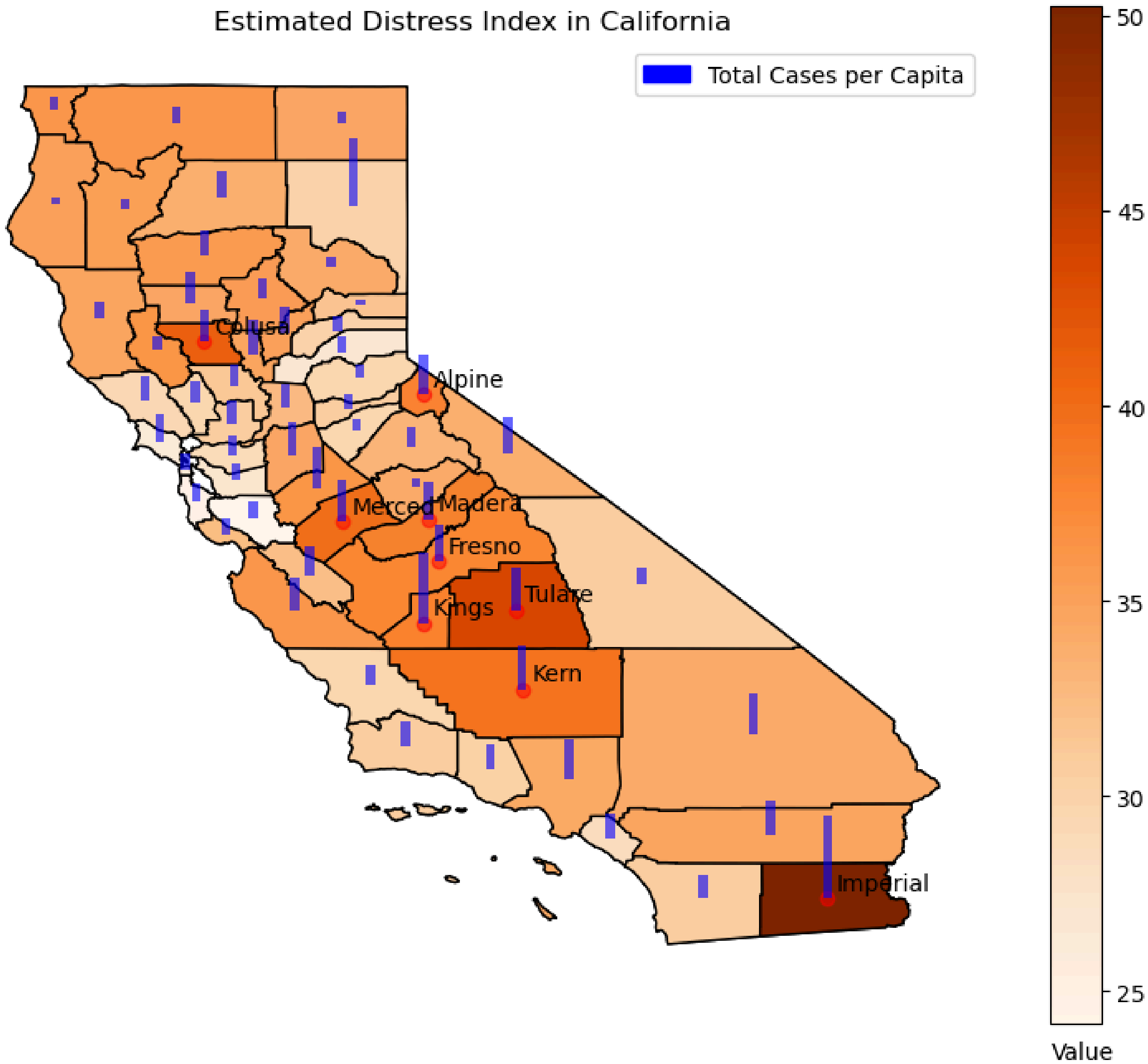
```
)

# Add text to let people know what we are plotting
gax.annotate('Percentage',xy=(0.65, 0.05),  xycoords='figure fraction')
gdf.head()
# Plot county names of high personal income per capita
gdf_highquantile = gdf[gdf['avg_percent_frequent_physical_distress'] > gdf['avg_per
gdf_highquantile.plot(ax=gax, color='red', alpha = 0.5)
for x, y, label in zip(gdf_highquantile['Coordinates'].x, gdf_highquantile['Coordin
    gax.annotate(label, xy=(x,y), xytext=(4,4), textcoords='offset points')


# I don't want the axis with long and lat
plt.axis('off')

# Add a title
plt.title('Average Percent Frequent Physical Distress by County in California')


plt.show()
```



This is a map highlighting my main x variable considered for exploration of moderation effects, physical distress. I highlight the areas with the most physical distress. We will look for correlation between the this and cases.

In [91]:

```
fig, gax = plt.subplots(figsize = (10,8))

# Plot the state level
state_df[state_df['NAME'] == 'California'].plot(ax = gax, edgecolor='black',color='

# Plot the counties and data
county_df_covid.plot(
    ax=gax, edgecolor='black', column='distress_index', legend=True, cmap='Oranges'
)

# Add text to let people know what we are plotting
gax.annotate('Value',xy=(0.67, 0.05),  xycoords='figure fraction')
gdf.head()
# Plot county names of high personal income per capita
gdf_highquantile = gdf[gdf['distress_index'] > gdf['distress_index'].quantile(0.85)
gdf_highquantile.plot(ax=gax, color='red', alpha = 0.5)
for x, y, label in zip(gdf_highquantile['Coordinates'].x, gdf_highquantile['Coordin
    gax.annotate(label, xy=(x,y), xytext=(4,4), textcoords='offset points')

# Add bars representing total cases per capita
for x, y, value in zip(gdf['Coordinates'].x, gdf['Coordinates'].y, gdf['total_cases
    gax.bar(x, value * 10, bottom=y, width=0.1, color='blue', alpha=0.6)

# Add a legend showing how distress_index is blue
blue_patch = mpatches.Patch(color='blue', label='Total Cases per Capita')
plt.legend(handles=[blue_patch])


# I don't want the axis with long and lat
plt.axis('off')

# Add a title
plt.title('Estimated Distress Index in California')

plt.show()
```



This is the ultimate map where I try to tie everything together. I combine mental, physical distress, and unemployment rate to create an index for total distress. Then plot cases as a bar on the same map so it can be easier to see the correlation between cases and distress. From this graph, I can somewhat see that areas with higher distress have higher cases on average. So, for my research question this indicates that physical distress may impact/ moderate the relationship between socioeconomic/ weather factors and cases

Conclusion

In this project, I hoped to add to existing research about weather, socioeconomic conditions, and COVID-19. To contribute to additional understanding about COVID-19, I run a regression to attempt to find out how physical and mental distress moderates the relationship between socioeconomic/weather variables and COVID-19 cases. I use a log linear regression model involving interaction terms to attempt to discover this effect. The result showed that physical distress primarily had a positive moderating effect, while mental distress had the opposite. In the future, I plan to consider implementing additional controls and using a machine learning to consider another perspective. Also, I plan to webscrape Reddit as a means to better identify levels of distress to enhance my data. Future controls could include dummies regarding overcrowding/ social distancing measures like how Shen et al. (2020) explored it. Satellite data would be an intriguing way to integrate this data in future research

In [96]:

```
# Word count in markdown cells
import json

with open('Project1Code.ipynb') as json_file:
    data = json.load(json_file)


wordCount = 0
for each in data['cells']:
    cellType = each['cell_type']
    if cellType == "markdown":
        content = each['source']
        for line in content:
            temp = [word for word in line.split() if "#" not in word]
            wordCount = wordCount + len(temp)

print(wordCount)
```

```
2064
```